KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

# HP Elite c640 14 inch G3

## Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
HP Elite c640 14 inch G3 Chromebook (Intel Core).

### In this guide


1 Read first: safety, tools, basic skills ..... 2
2 Know the machine: specs, anatomy, parts list ..... 9
3 Triage: symptom → repair finder ..... 11
4 Software repairs (S1–S4) ..... 12
5 Hardware repairs (R1–R17) ..... 17
6 Out of scope: send to HP / lead tech ..... 42
7 Sources ..... 43



Rev 1.0 • October 2026
Built from HP's official Maintenance and Service Guide for this model.
Part numbers: verify on partsurfer.hp.com before ordering.

# 1 Read first

This guide covers every HP Elite c640 14 inch G3 Chromebook repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

**How this guide was built.** Every screw count, screw size and part number here comes from HP's official Maintenance and Service Guide for the Elite c640 14 inch G3 Chromebook. This guide rewrites HP's procedures step by step for a student bench and adds tests and checks.

**Keep the gold strip and silver foil.** HP's guide says the gold strip and silver foil inside the bottom cover must stay intact. Don't peel them; they ground the cover.

## Bench levels

| Level | Who may do it | Typical repairs |
|---|---|---|
| LEVEL 1 | Any trained student | Software fixes, single keys, bottom cover |
| LEVEL 2 | Student with supervisor sign-off on this repair | Battery, Wi-Fi, touchpad, speakers, USB board, display parts, hinges |
| LEVEL 3 | Lead tech or adult supervisor only | Fan/heat sink, top cover (keyboard) transplant, liquid damage, system board |

## Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under HP warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

## Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

## Tools and supplies

| Item | Use | Notes |
|---|---|---|
| Phillips #1 (magnetic) and #0 | All screws | HP lists a magnetic Phillips P1 |
| Tweezers | Stretch-release panel tape | Pull slowly and flat |
| Plastic spudger, opening picks, plastic tweezers | Clips, latches, ribbons | Only plastic touches connectors |
| ESD mat + wrist strap | Static protection | |
| Magnetic screw mat | Keeping screws in order | Screws are often different lengths |
| 90%+ isopropyl alcohol + soft brush | Cleaning contacts, spill residue | |
| Kapton tape | Re-taping cables and connectors | |
| Recovery USB or SD card, 8 GB+ | ChromeOS recovery | Chromebook Recovery Utility |
| Known-good USB-C charger | Testing power and charging | |
| Safety glasses | Cracked-screen removal | |
| Fire-safe battery bin or sand bucket | Damaged batteries | Required at every bench |

# Basic skills (start here if you're new)

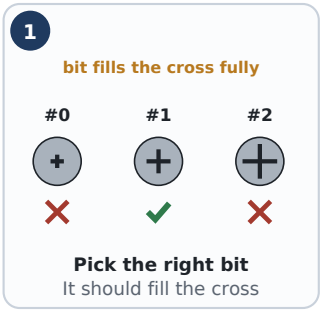
Every repair in this guide uses the same handful of skills. Learn these six first and the rest of the guide is just *which screws* and *which cable*. If a word in a repair is unfamiliar, look it up in the word list at the end of this chapter. Hard words in the steps are also explained in grey the first time they appear.

## How a repair page works

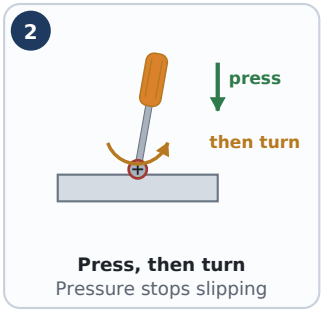
| Part of the page | What it tells you |
|---|---|
| Coloured bar with the code (R5) and LEVEL | What the repair is and who may do it. Level 1: any trained student. Level 2: with a supervisor's sign-off. Level 3: lead tech only. |
| Fixes / Part / Time / Tools | Read this first. If you don't have the part or the tools, stop and get them before you open anything. |
| **In plain words** | One or two sentences on what you're about to do and why. |
| At a glance (numbered pictures) | The idea of the repair in four pictures. They show the idea, not your exact model. |
| Where it is on this model | A map of the inside with the part highlighted. |
| Step by step (☐ boxes) | The exact steps. Tick each box as you go so you never lose your place. |
| **Common mistakes** / If something goes wrong | The traps other students fall into, and what to do when a step doesn't work. |
| Verify before closing the ticket | The checks that prove the repair worked. Never skip them. |
| First-teardown notes | Write in what you found the first time (real screw counts, tape positions). It helps the next student. |

## Skill 1: Screws

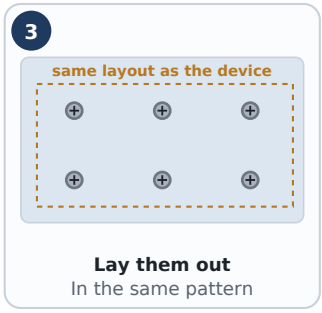
Most damage on a repair bench comes from screws: stripped heads, screws in the wrong hole, or a long screw driven through a board. Go slowly; screws are tiny and soft.



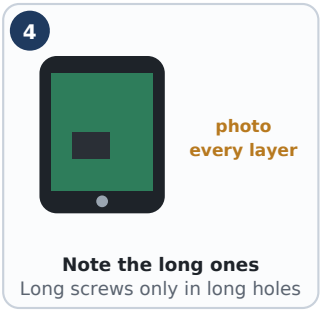

**Pick the right bit**
It should fill the cross




**Press, then turn**
Pressure stops slipping




**Lay them out**
In the same pattern

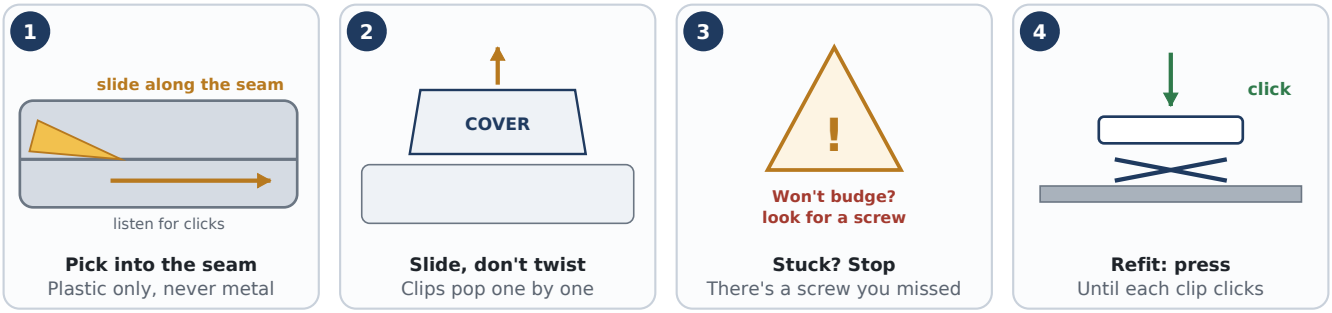



**Note the long ones**
Long screws only in long holes

- Most Chromebook screws take a **Phillips #0 or #1** bit. If the bit wobbles or slips, try the other size.
- Turn **left to loosen** (counter-clockwise), right to tighten. Tighten until it stops, then stop. "Snug" is enough: over-tightening cracks plastic.
- **Captive screws** stay in the cover. Loosen them until they spin freely; they won't come out.
- If a screw head starts to strip (the cross rounds off), stop and get the lead tech. Don't keep turning.

## Skill 2: Opening clips

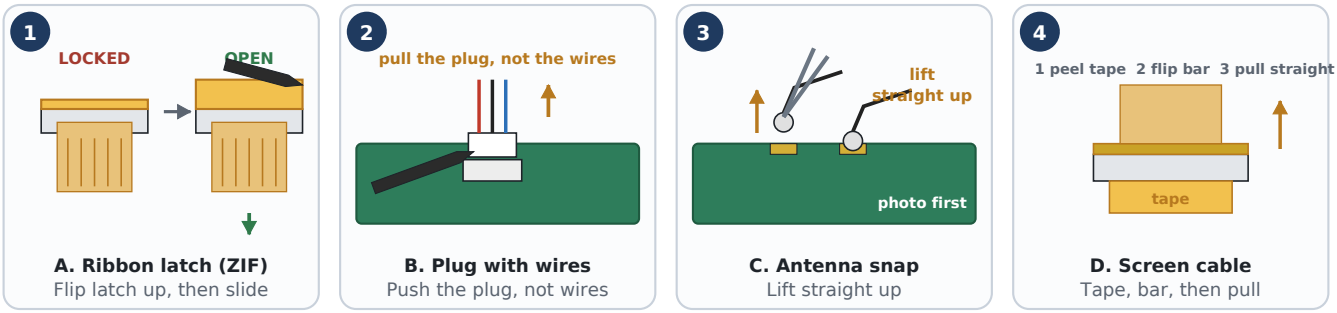
Covers and bezels are held by small plastic clips as well as screws. You release them by sliding a plastic pick along the seam, not by pulling.




- A click or pop is normal. A crack or creak means stop and look for a screw.
- Start where the guide says (usually a corner) and work around slowly.
- Never pry near the battery or against the screen glass.

## Skill 3: The four kinds of connectors

Inside every Chromebook, parts plug into the main board in only four ways. Learn these and you can unplug anything safely.




| Type | Looks like | How to unplug | How to plug back in |
|---|---|---|---|
| A. Ribbon latch (ZIF) | Flat, thin, see-through cable with gold stripes, in a slot with a small hinged flap | Flip the flap up with a fingernail or spudger. The ribbon slides out with no force. | Gold stripes facing the same way as before. Push it in straight and all the way, then close the flap. |
| B. Plug with wires | Small white plug with coloured wires (battery, speakers, fan) | Push the plug body out with a spudger, gently rocking. Never pull the wires. | Line it up, press straight in until it seats flat. |
| C. Antenna snap | Tiny round gold button on a thin black or white wire | Lift straight up with plastic tweezers or a fingernail. | Hold it right over the post and press straight down until it snaps. |
| D. Screen cable | Wider connector, often under tape, with a metal bar or pull tab | Peel the tape, flip up the metal bar (or pull the tab), then pull straight out. | Push straight in, flip the bar down, put the tape back. |

**Battery first, battery last.** Unplug the battery (type B) before you unplug anything else, and plug it back in only after everything else is connected.

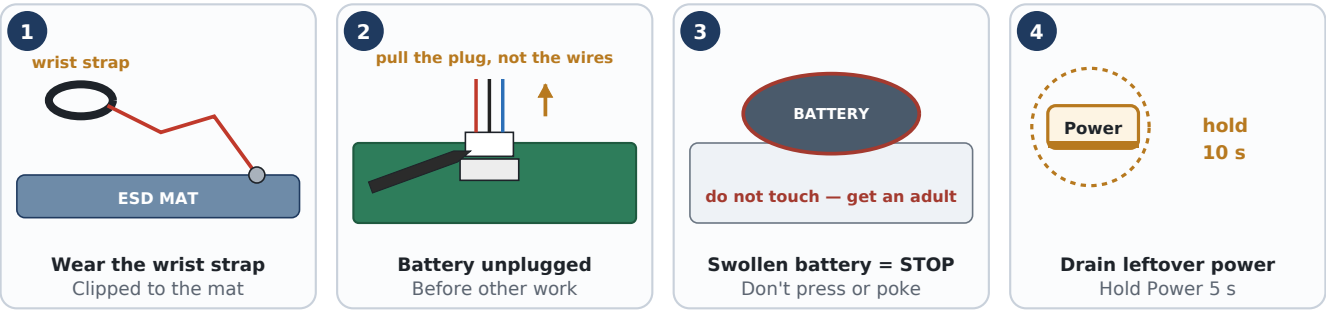
## Skill 4: Tape, Mylar and adhesive

Many parts are held by tape as well as screws. Some tape is meant to be reused; some must be replaced.




- **Mylar** is thin plastic sheet that insulates parts. Put it back where it was: it stops short circuits.
- **Kapton** is thin amber heat-proof tape for holding cables down. It peels off and can be reused if clean.
- **Stretch-release tape** (under screens and batteries) lets go when you pull its tab slowly and flat. Pulling up snaps it.
- Conductive (shiny) tape on touchpads and cables must touch metal again when you refit, or the part acts strangely.

## Skill 5: Static and battery safety




- Static from your body can kill a board without a spark you can see. Strap on, and touch boards only by the edges.
- A swollen, hot, hissing or sweet-smelling battery is an emergency: don't press it, move it to the fire-safe bin with an adult.
- Water, drinks and food stay off the bench.

## Skill 6: Photograph, then test before you close




- Your photos are your instructions for putting it back together. Take more than you think you need.
- Plug in, power on and test the repaired part while the cover is still off. Finding a loose cable now saves taking it all apart again.
- Count your screws at the end. A leftover screw means something isn't fastened.

## Word list

| Word | What it means |
|---|---|
| **Bezel** | The plastic frame around the screen. |
| **Bottom cover (base, D cover)** | The panel underneath the Chromebook. |
| **Top cover (palm rest, C cover)** | The part your wrists rest on; often has the keyboard built in. |
| **Lid / A cover** | The back of the screen half. |
| **Board / system board / mainboard** | The main circuit board with the processor, memory and storage soldered on. |
| **Daughterboard / I/O board** | A small extra board holding some ports, joined to the main board by a ribbon cable. |
| **Ribbon cable (FFC)** | A flat, thin cable with gold stripes at the ends. |
| **ZIF connector** | A slot with a small flip-up latch that clamps a ribbon cable. "Zero insertion force": open it and the ribbon slides in or out freely. |
| **eDP / LVDS cable** | The video cable that carries the picture from the board to the screen. |
| **Coax / antenna lead** | The thin round wires from the Wi-Fi card to the antennas in the lid. |
| **Spudger** | A plastic or nylon pry stick. Safe to use near electronics. |
| **Pick** | A thin plastic triangle used to open seams and clips. |
| **Captive screw** | A screw that stays in the part after you loosen it. |
| **Broad-head screw** | A screw with a wide, flat head. Often used where it clamps plastic. |
| **Mylar** | Thin plastic insulating sheet. |
| **Kapton tape** | Thin amber heat-proof tape. |
| **Thermal paste / pad** | Grey paste or a soft pad that carries heat from the processor to the heat sink. |
| **Heat sink** | Copper or aluminium part that pulls heat off the processor. |

| Word | What it means |
|---|---|
| **ESD** | Electrostatic discharge: static electricity that can damage boards. |
| **Isolator** | A rubber piece that stops a part (usually a speaker) from rattling. |
| **Powerwash** | A factory reset of ChromeOS. Erases files saved on the device. |
| **Recovery** | Reinstalling ChromeOS from a USB stick or SD card. |
| **EC reset (hard reset)** | Restarting the small chip that controls power, charging and the keyboard. |
| **Enroll** | Connecting the Chromebook to the district's management so school policies apply. |
| **eMMC** | The storage chip on the board (like a small SSD). Soldered, so it can't be swapped. |
| **Level 1 / 2 / 3** | Who may do a repair: any trained student / student with sign-off / lead tech only. |

# 2 Know the machine

## Specs that matter for repair

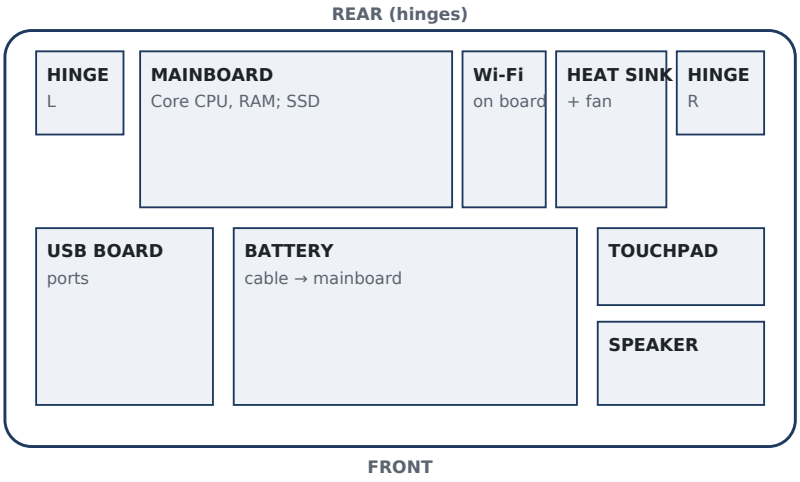
| Item | Elite c640 G3 | Why it matters |
|---|---|---|
| Processor | Intel Core (11th gen), soldered | System board level |
| Memory / storage | Soldered RAM; SSD on the system board | Move the SSD to a new board |
| Display | 14" FHD, glued bezel, tape-mounted panel | Bezel and panel need new adhesive |
| Battery | HP M12585-005 | 5 screws |
| Access | Bottom cover: 6 screws | Pry, then lift |
| Wireless | No separate Wi-Fi spare in HP's guide | Treat as system-board level |
| Cooling | Fan + heat sink | Clean and re-paste every time |

## How it comes apart

A screwed-on bottom cover comes off first. Under it: battery, system board, fan, heat sink, USB board, speakers and touchpad. The keyboard is built into the top cover, which is the last part left after everything else is removed.

## Inside the base

With the cover off you see the parts below. It's a rough map; correct it on your first teardown.




*ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.*

## Elite c640 G3 parts list

HP spare part numbers from the official Maintenance and Service Guide. Confirm on partsurfer.hp.com by serial number before ordering.

| Part | Part number | Repair |
|---|---|---|
| Bottom cover | N12891-001 | R2 |
| Battery | M12585-005 | R3 |
| Touchpad / cable | N13682-001 / N14811-001 | R5 |
| Speakers (pair) | N12919-001 | R6 |
| USB board / cable | N13679-001 / N13681-001 | R7 |
| Fan / heat sink | N12918-001 / N13683-001 | R8 |
| Display bezel / adhesive kit | N12925-001 / M17753-001 | R10 |
| FHD panel / tape kit | N13685-001 / N13937-001 | R11 |
| Display cable | N12926-001 | R12 |
| Webcam module / cable | N12929-001 / N10434-001 | R13 |
| Hinges / hinge cover | N12927-001 / N12928-001 | R14 |
| Fingerprint reader / board | N14814-001 / N12915-001 | R15 |
| Top cover with keyboard | N14927-xxx (by country) | R15 |
| System board | N13676-001 (example) | R17 |

# 3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

| Symptom | Try first | Then | Part likely needed |
|---|---|---|---|
| Won't turn on, no lights | Known-good charger 30 min → S1 | R4 → R3 | Battery, or system board |
| Not charging | Other USB-C port and charger → S1 | R3 → Sec. 6 if a port is dead | Battery or system board |
| Battery drains fast / low health in S2 | S2 battery test | R3 | Battery |
| Swollen battery / bulging case | STOP — Level 3 | R3 by lead tech | Battery |
| Black screen, power light on | External display via USB-C; S1 | R4 → R12 → R11 | Display cable or panel |
| Flicker or lines that change with lid angle | S1 | R12 | Display cable |
| Cracked or blotchy screen | — | R11 | LCD panel |
| Bezel loose or broken | Reseat clips | R10 | Bezel |
| Lid floppy, hinge cracked | Check hinge screws | R14 | Hinge(s) |
| One key missing or broken | — | R1 | Keycap |
| Several keys or a row dead | S2 keyboard test; S1; S4 | R4 → R15 | Keyboard / top cover |
| Keys type wrong symbols | S4 | S4 | None |
| Touchpad dead or erratic | S1; check touchpad settings | R4 → R5 | Touchpad or its cable |
| USB-A / SD card / audio dead | Try another device in the port; S1 | R7 | I/O board or its cable |
| Camera not detected | Privacy shutter open? S1 | R13 | Camera module |
| No sound (headphones work) | Volume, output; S1; S3 | R6 | Speakers |
| Runs hot, fan loud, shuts down | Check vents for lint; S2 | R8 | Fan or heat sink |
| Liquid spill | Power off NOW, unplug | R16 | Depends on damage |
| "ChromeOS is missing or damaged" | S3 recovery | R17 | System board (storage) |
| Cracked bottom or lid shell | — | R2 | Cover |

# 4 Software repairs

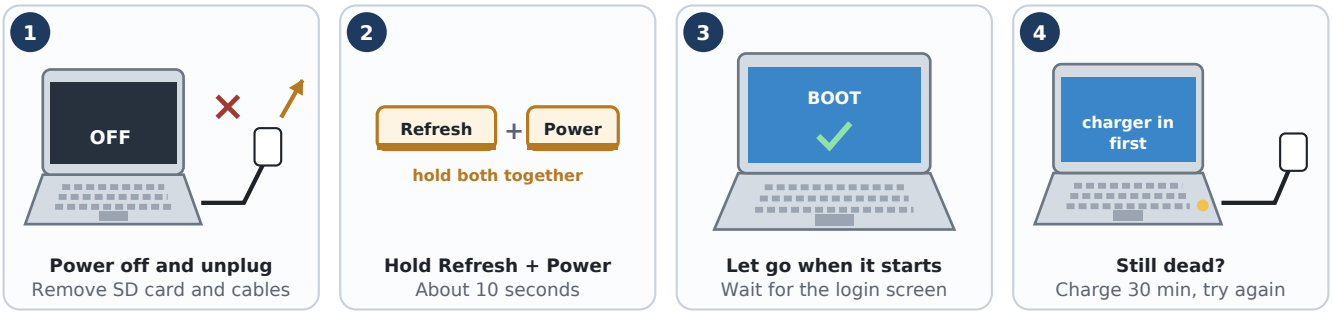
No tools needed. Always do S1 before opening a device.

## S1 Hard reset (EC reset) LEVEL 1

| | |
|---|---|
| **Fixes** | Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad |
| **Time** | 2 minutes |
| **Tools** | Known-good USB-C charger |
| **Data loss** | None |

**In plain words.** A hard reset restarts the small chip that runs power, charging and the keyboard. It's like unplugging something and plugging it back in, and it fixes a surprising number of "dead" Chromebooks. Nothing is erased.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

### Step by step

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

**Common mistakes**
- ✗ Giving up after one try without the charger.
- ✗ Holding only Power (that's just a shutdown).
- ✗ Using a charger you haven't tested.

**If something goes wrong**
**Nothing at all, even charging** → Try the other USB-C port, then check the battery (battery repair).
**Starts then shuts off** → Run diagnostics (S2) for the battery.

### Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

**Tip.** After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

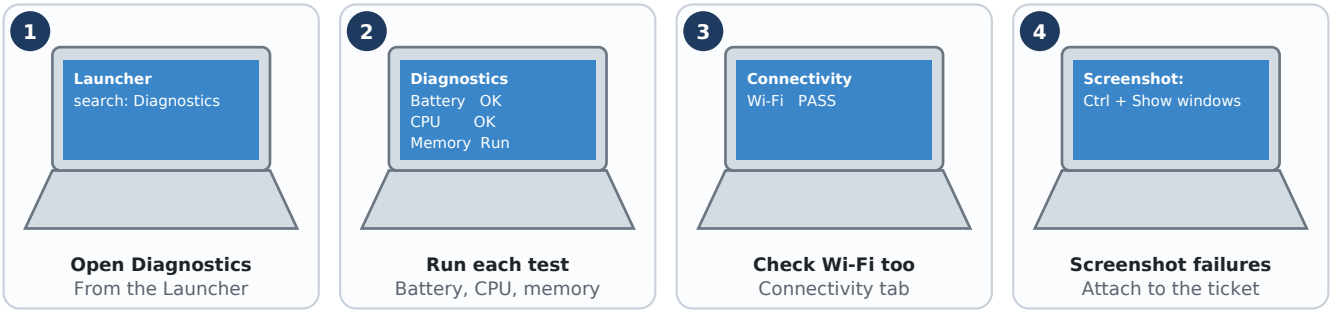
## S2 Built-in hardware diagnostics

LEVEL 1

| | |
|---|---|
| **Fixes** | Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi |
| **Time** | 5–15 minutes |
| **Tools** | None |
| **Data loss** | None |

**In plain words.** Chromebooks have a built-in checkup app. It tests the battery, processor, memory and more, and tells you what's failing before you order anything.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

### Step by step

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

**Common mistakes**

✗ Skipping the memory test because it's slow.
✗ Not saving the results to the ticket.
✗ Replacing a battery that tests healthy.

**If something goes wrong**

**Battery health low** → Battery repair.
**Memory or storage fails** → System board: Section 6 or lead tech.

### Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

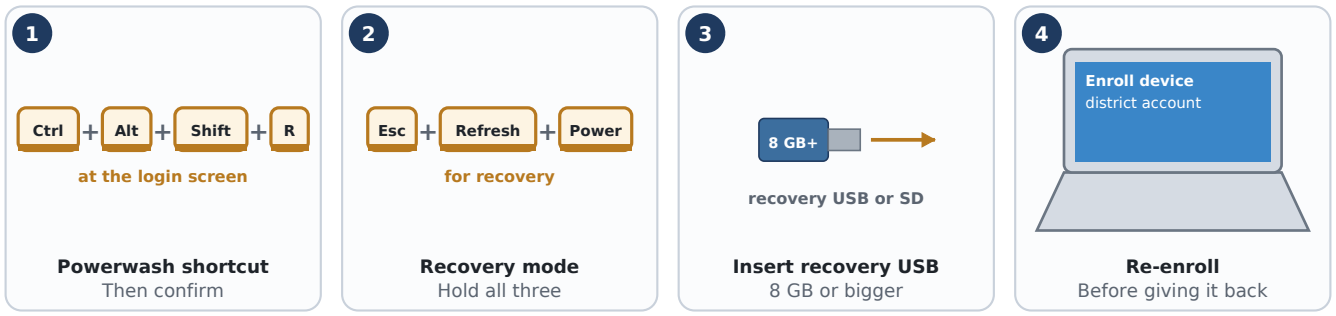
# S3 Powerwash and USB recovery

LEVEL 1

| | |
|---|---|
| **Fixes** | Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update |
| **Time** | Powerwash 10 min; recovery 20–40 min |
| **Tools** | Recovery USB or SD card (8 GB+) |
| **Data loss** | **Yes.** Erases all local files (Downloads). |

**In plain words.** A Powerwash wipes the Chromebook back to factory settings, like a fresh start. Recovery goes further and reinstalls ChromeOS from a USB stick. Both erase files saved on the device, so warn the student first.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

**District-managed devices.** Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

## Step by step

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash (factory reset):** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **HP Elite c640 14 inch G3 Chromebook** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

**Common mistakes**
✗ Not warning that Downloads will be erased.
✗ Choosing Developer Mode (never on district devices).
✗ Handing it back before re-enrolling.

**If something goes wrong**
**"ChromeOS is missing or damaged" again after recovery** → Storage failing: system board, Section 6.
**Can't enroll** → Lead tech checks the Admin console.

## Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.

- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

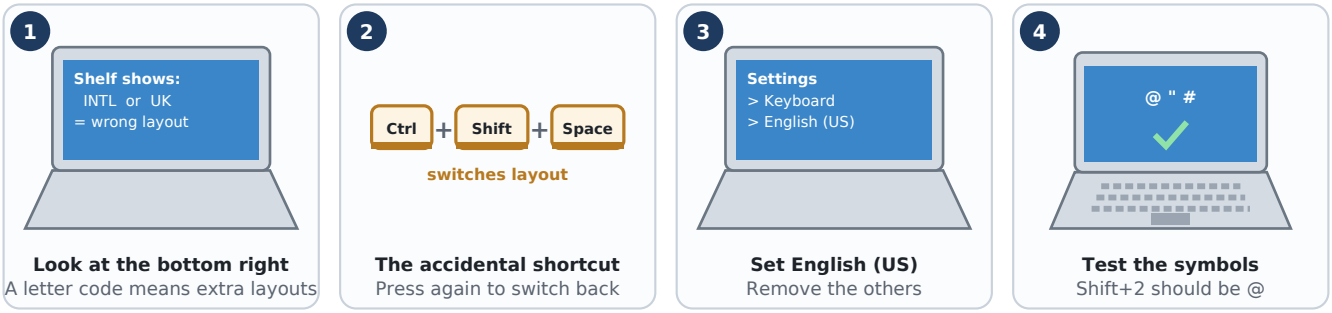
## S4 Keyboard layout and international keyboard fix

LEVEL 1

| | |
|---|---|
| **Fixes** | Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout |
| **Time** | 5 minutes |
| **Tools** | None |
| **Data loss** | None |

**In plain words.** If keys type the wrong symbols, the keyboard usually isn't broken: the computer has been switched to a different keyboard language. A key combination switches it by accident.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

### Step by step

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

**Common mistakes**

✗ Ordering a keyboard for a software problem.
✗ Leaving US International installed.
✗ Changing it when the district sets it by policy.

**If something goes wrong**

**Wrong layout comes back** → District policy sets it: tell the lead tech.
**Printed keys are a different language** → Wrong replacement part: order the US one.

## Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

# 5 Hardware repairs

Most repairs start with the bottom cover (R2) and disconnecting the battery (R3). Screw counts and sizes come from HP's guide.

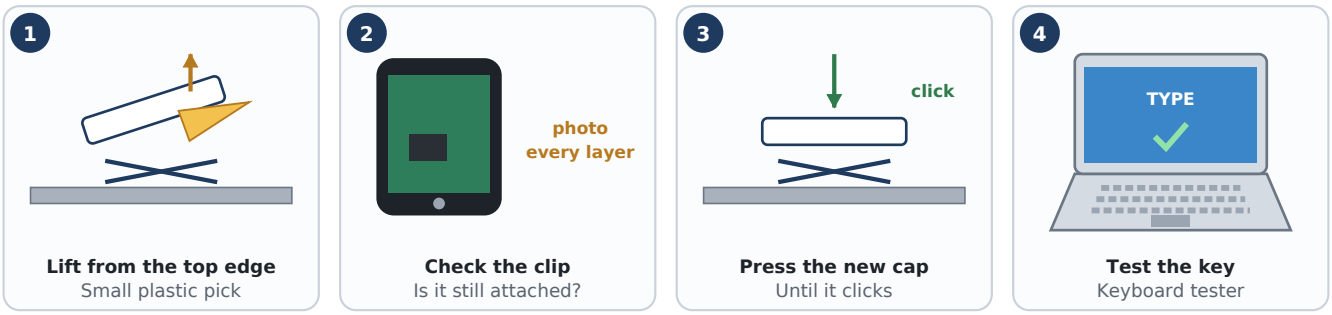
**Golden rules.** Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

## R1 Single keycap replacement — LEVEL 1

| | |
|---|---|
| **Fixes** | Missing, cracked or loose key |
| **Part** | Individual key (ask the manufacturer or harvest from a scrapped keyboard) |
| **Time** | 5 minutes |
| **Tools** | Plastic pick, tweezers |

**In plain words.** A key is a plastic cap clipped onto a small scissor-shaped hinge. A missing or broken key can usually be fixed with just the cap, in a few minutes.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

### Where it is on this model

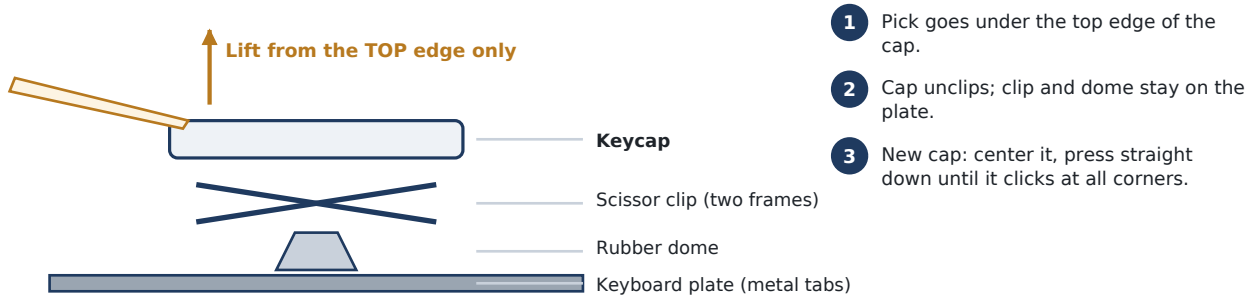



*Illustration: side view of one key. Not to scale.*

### Step by step

**1.** ☐ Power off. Open the lid flat on the mat.

2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.
3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

**Common mistakes**
✗ Prying from the bottom edge (snaps the clip).
✗ Pressing a cap on crooked.
✗ Using a cap from a different model.

**If something goes wrong**
**Clip is broken** → Order the key with its clip, or replace the keyboard.
**Key types nothing** → Keyboard ribbon or keyboard: connectors repair first.

### Verify before closing the ticket

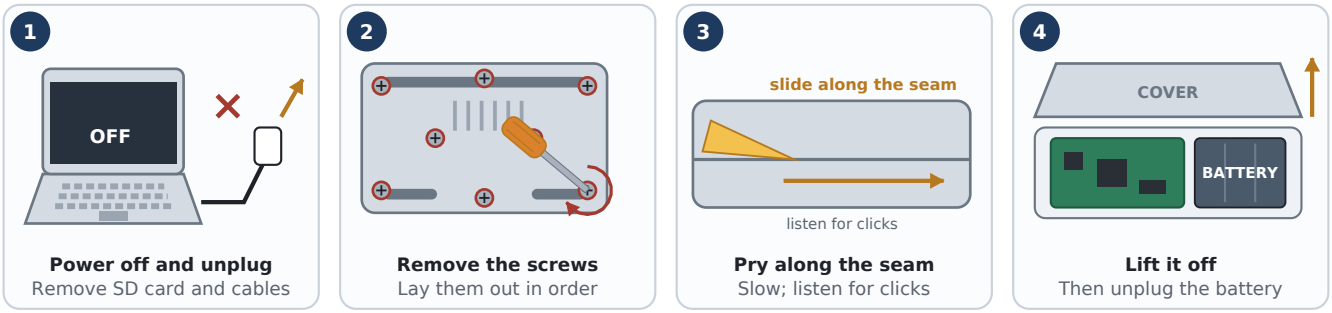
- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

## R2 Base cover (bottom cover) removal — LEVEL 1

| | |
|---|---|
| **Fixes** | Cracked or missing bottom cover; the first step for most internal repairs |
| **Part** | Bottom cover N12891-001 |
| **Time** | 10 minutes |
| **Tools** | Phillips driver, plastic picks, screw mat |

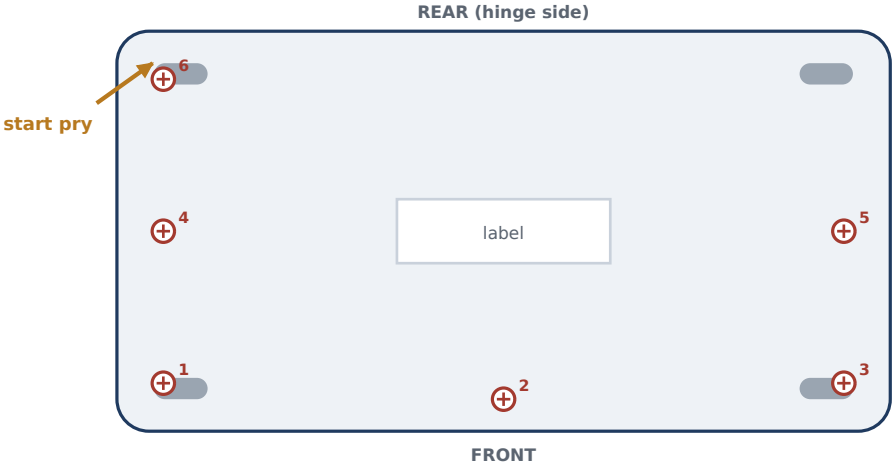
**In plain words.** The bottom cover is the plastic panel underneath. It's held by small screws and plastic clips. Taking it off is the first step for almost every repair inside.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Remove the 6 bottom screws.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

*Illustration: underside. Screw count from HP's guide; positions illustrative.*

## Step by step

1. ☐ Shut down fully, unplug, remove any SD card.
2. ☐ Close the lid and turn the computer upside down on the mat, hinges away from you.
3. ☐ Remove the 6 screws: 4 Phillips M2.5×6.0 along the top (hinge) edge and 2 Phillips M2.0×4.0 along the front edge.
4. ☐ Release the cover with a plastic pick at the rear edge, then work around the sides. Lift it off.
5. ☐ **Disconnect the battery now** (battery repair, first steps) before anything else.
6. ☐ To refit: line up the cover, press around the edges until every clip snaps, then refit every screw. Snug only. Check the gold strip and silver foil are still in place.

**Common mistakes**

✗ Missing a screw under a rubber foot and cracking the cover.
✗ Mixing up long and short screws.
✗ Using a metal tool in the seam (scratches, cracks).

**If something goes wrong**

**Cover won't release in one spot** → There's still a screw there. Check under the feet and labels.
**A clip broke** → It usually still holds. If it rattles, order a new cover.

## Verify before closing the ticket

- No gaps around the seam; device sits flat.
- Every screw refitted, none left over.

**First-teardown notes** — fill in the first time you do this repair

Which holes take the longer screws? _____

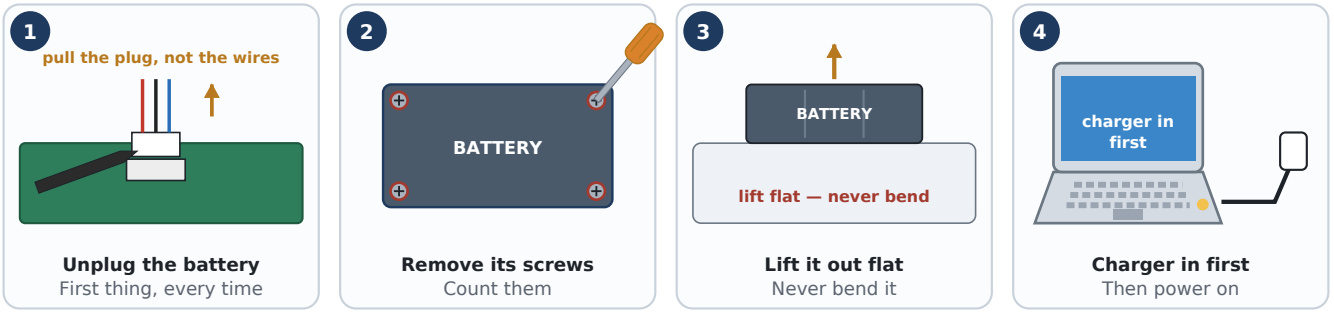
# R3 Battery replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Won't hold charge, low health in S2, swollen battery |
| **Part** | Battery M12585-005 |
| **Time** | 15–20 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap |

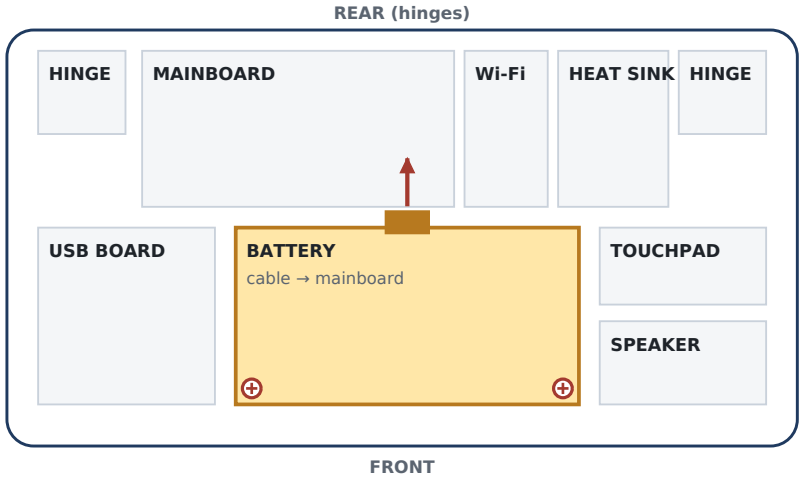
**In plain words.** The battery is a flat pack that powers the Chromebook when it's unplugged. To replace it you unplug its cable from the board, remove a few screws, and lift it out. Batteries can be dangerous if bent or punctured, so handle them gently.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

*Illustration: base interior, battery highlighted. Screw positions illustrative.*

**Swollen battery = Level 3.** Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

## Step by step

**1.** ☐ Remove the bottom cover.

2. ☐ Lift any tape over the battery connector and disconnect the battery cable from the system board.
3. ☐ Remove the 5 Phillips M2.0×4.0 screws.
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit the screws, reconnect the cable.
6. ☐ Refit the bottom cover. Plug in the charger before powering on.

**Common mistakes**
✗ Pulling the battery cable by its wires.
✗ Prying under the battery with a metal tool.
✗ Thinking it's dead because it won't start without the charger.

**If something goes wrong**
**Swollen, hot, hissing or smells sweet** → STOP. Don't press it. Fire-safe bin, get an adult.
**Won't power on with new battery** → Plug in the charger for 30 minutes, then hard reset (S1).

### Verify before closing the ticket

- Charging light comes on; battery percentage shows.
- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

**HP battery notes.** HP says to use only HP-supplied batteries. If a battery is swollen or stuck, stop and contact HP support rather than forcing it.

## R4 Reseating internal connectors — LEVEL 2

| | |
|---|---|
| **Fixes** | Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop |
| **Part** | None, unless a cable is damaged |
| **Time** | 15–25 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol |

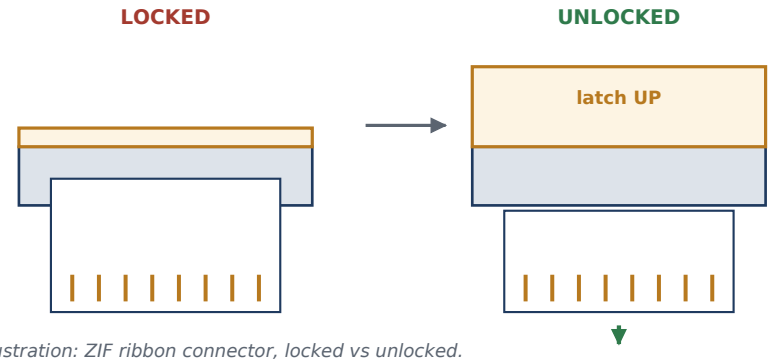
**In plain words.** Inside, parts connect to the main board with small plugs and flat ribbon cables. A drop can wiggle one loose. Unplugging it, checking it, and plugging it back in properly fixes many problems for free.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

*Illustration: ZIF ribbon connector, locked vs unlocked.*

| Connector | Type | How to release |
|---|---|---|
| Battery cable | Board-to-wire | Lift tape; pry at the connector body |
| Keyboard ribbon | ZIF ribbon | Unlock the latch; slide out |
| Touchpad ribbon | ZIF ribbon | Unlock the latch; slide out |
| Display (eDP) cable | Locking connector | Lift tape and the locking bar, pull straight |
| USB board cable | ZIF ribbon(s) | Unlock both ends first |
| Speaker cable | Board-to-wire | Pry at the plug body |
| Fan cable | Board-to-wire | Pry at the plug body |
| Fingerprint reader cable | ZIF ribbon | Unlock the latch; slide out |

## Step by step

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.
6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

**Common mistakes**

✗ Pulling a ribbon without unlocking the latch.
✗ Inserting a ribbon crooked or only half way.
✗ Forgetting to reconnect the battery last.

**If something goes wrong**

**Ribbon is creased or torn** → Replace that cable or part.
**Symptom comes back later** → The part itself is failing: replace it.

## Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

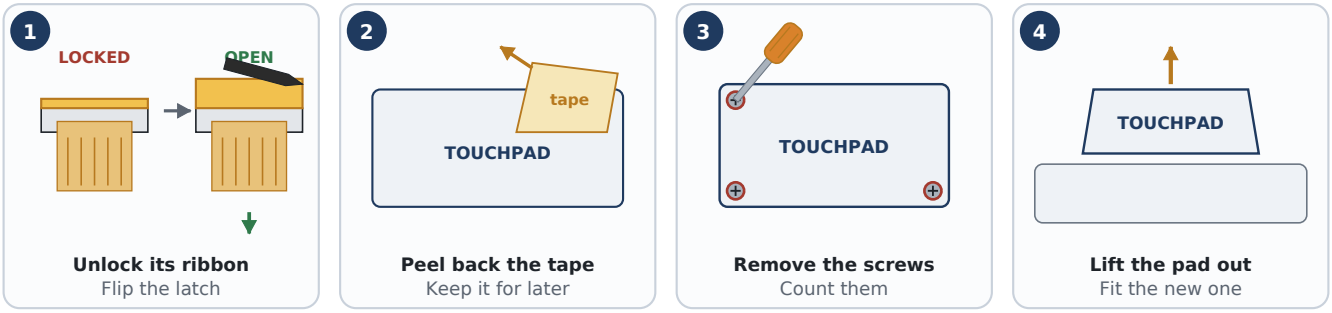
# R5 Touchpad and touchpad cable replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Touchpad dead, erratic, cracked or won't click (keyboard works) |
| **Part** | Touchpad N13682-001; touchpad cable N14811-001 |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, Kapton tape |

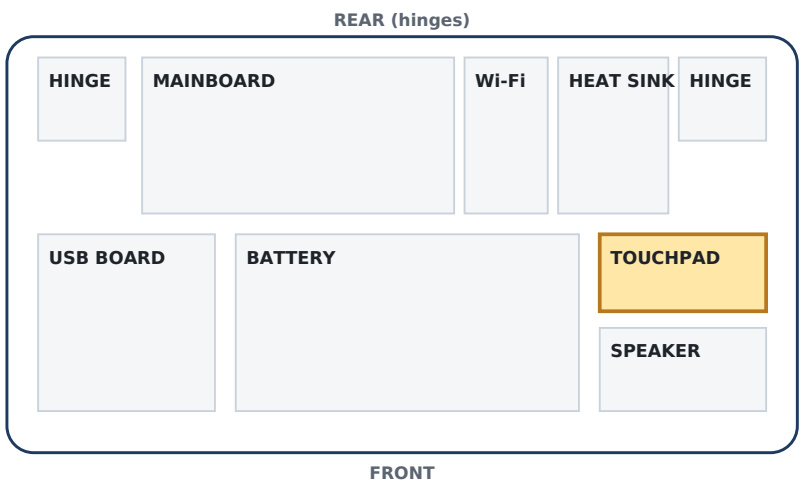
**In plain words.** The touchpad is a flat glass or plastic pad held from underneath by a few small screws and a bracket. A ribbon cable connects it to the board.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

*Illustration: base interior, touchpad highlighted. Position illustrative.*

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

## Step by step

1. ☐ Remove the bottom cover and the battery.
2. ☐ Lift the Mylar (a thin plastic sheet) over the touchpad connector and disconnect the touchpad cable (ZIF (a flip-up latch that clamps a flat ribbon cable)).
3. ☐ Remove the 3 Phillips M2.0×2.0 screws.

4. ☐ Lift the touchpad out. The cable is a separate part (N14811-001): transfer or replace it.
5. ☐ Fit the new pad, refit the screws and Mylar, reconnect.

**Common mistakes**
✗ Forgetting the conductive tape (the pad acts strange).
✗ Over-tightening screws (click stops working).
✗ Ribbon not locked.

**If something goes wrong**
**Click doesn't work** → Loosen the bracket screws a little.
**Cursor jumps around** → Check the tape and ribbon; try S3.

### Verify before closing the ticket

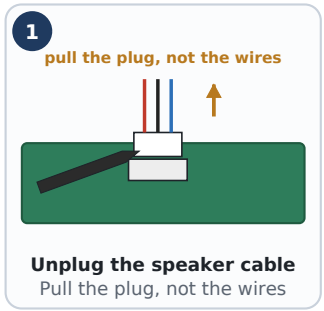
- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

## R6 Speaker replacement — LEVEL 2

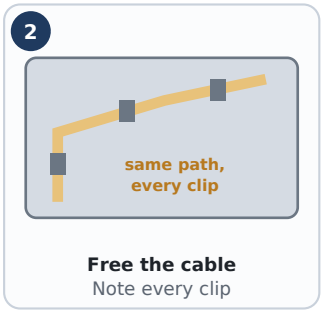
| | |
|---|---|
| **Fixes** | No sound or crackling from the speakers, with headphones working |
| **Part** | Speakers N12919-001 (pair) |
| **Time** | 20 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |

**In plain words.** The speakers are two small boxes joined by one cable. They unplug from the board, unroute from a few clips, and lift out. The rubber pieces stop them rattling, so keep them.

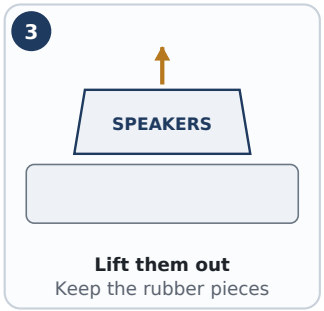
### At a glance




**Unplug the speaker cable**
Pull the plug, not the wires




**Free the cable**
Note every clip




**Lift them out**
Keep the rubber pieces




**Test sound**
Full volume, no rattle

The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

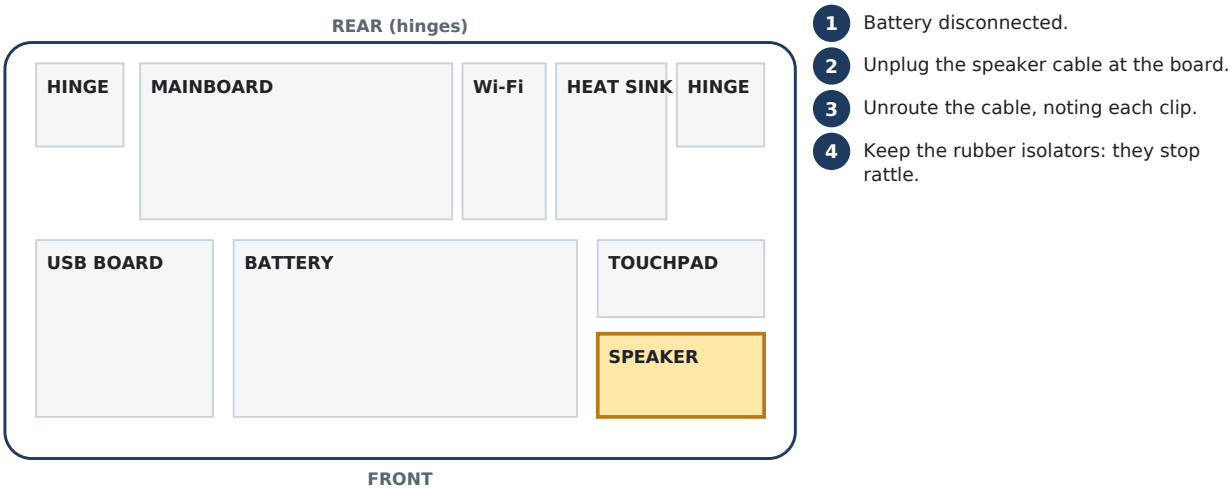



*Illustration: base interior, speaker highlighted. Position illustrative.*

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

## Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the speaker cable from the system board and free it from its routing clips.
3. ☐ Remove the 4 Phillips M2.0×5.0 screws (2 per speaker). Keep the rubber washers.
4. ☐ Lift both speakers out together.
5. ☐ Fit the new pair with the washers, route the cable and reconnect.

**Common mistakes**

✗ Losing the rubber isolators (speakers buzz).
✗ Pinching the cable under the cover.
✗ Replacing speakers when headphones are silent too.

**If something goes wrong**

**Headphones also silent** → Software or board: S1, S3, then lead tech.
**Rattle at high volume** → Missing isolator or loose screw.

## Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

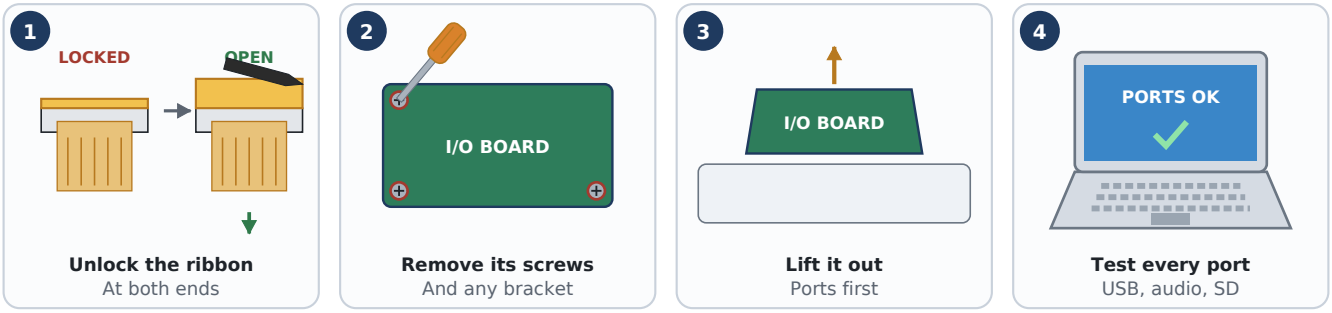
# R7 I/O (USB) daughterboard replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Dead USB-A, audio jack with the system board working |
| **Part** | USB board N13679-001; cable N13681-001 |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |

**In plain words.** The I/O board is a small separate board that holds some of the ports (like the USB-A port or headphone jack). It connects to the main board with a ribbon cable, so it can be swapped by itself.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

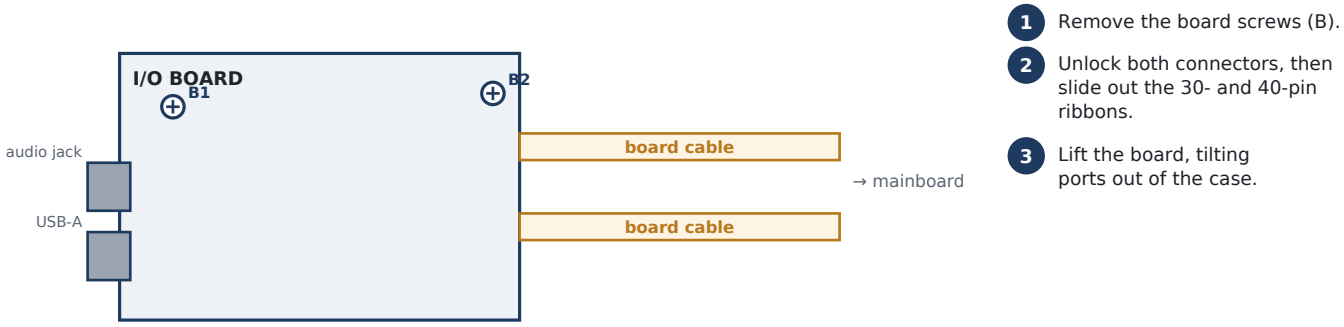



*Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.*

## Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Remove the bracket (2 Phillips M2.0×3.0) and the cable clip (1 Phillips M2.0×2.5).
3. ☐ Disconnect the USB board cable at both ends.
4. ☐ Remove the 3 Phillips M2.0×2.5 screws and lift the board out.
5. ☐ Fit the new board and its cable (N13681-001), refit the clip and bracket.

**Common mistakes**
✗ Ribbon in upside down or crooked.
✗ Forgetting the bracket.
✗ Testing only one port.

**If something goes wrong**
**Ports still dead** → Ribbon cable or the main board connector.
**USB-C dead** → Usually on the main board: Section 6.

### Verify before closing the ticket

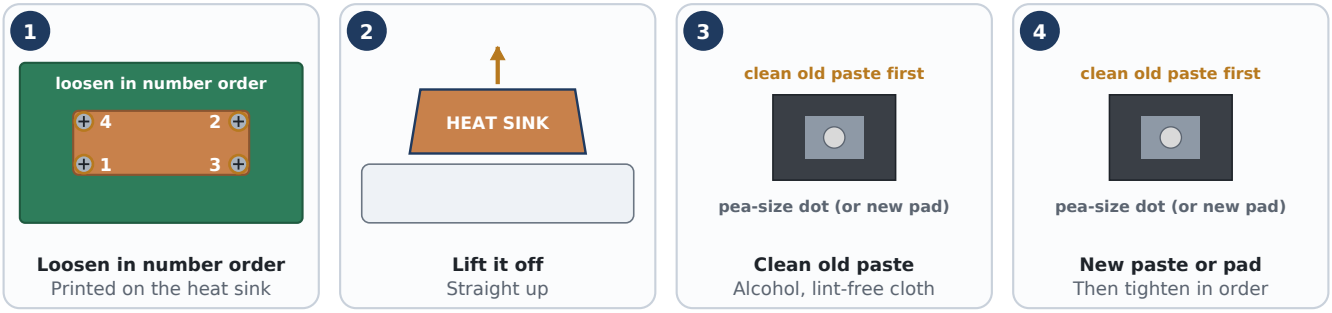
- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

## R8 Fan and heat sink replacement — LEVEL 3

| | |
|---|---|
| **Fixes** | Overheating, thermal shutdowns, loud or rattling fan |
| **Part** | Fan N12918-001; heat sink N13683-001 |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth |

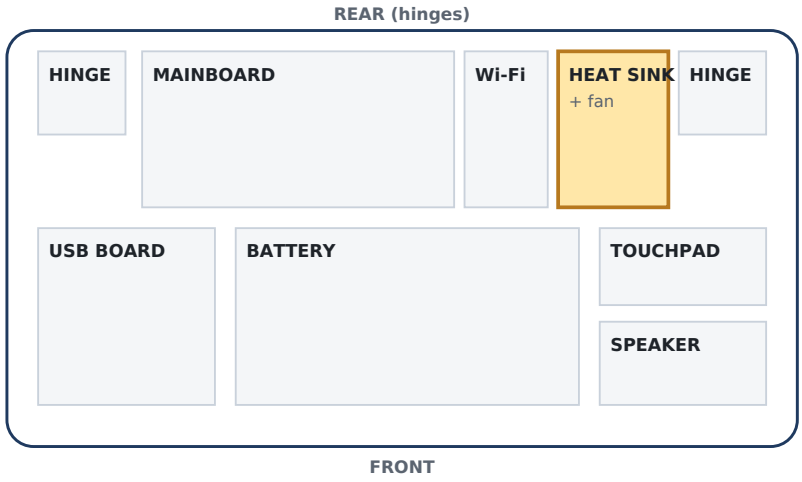
**In plain words.** The heat sink is a copper plate that pulls heat off the processor. A thin layer of thermal paste between them helps heat move. Whenever the heat sink comes off, the old paste must be cleaned off and new paste put on.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

*Illustration: base interior, cooling highlighted. Position illustrative.*

**Thermal material.** Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

## Step by step

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the fan cable from the system board and remove the 3 Phillips M2.0×3.0 screws holding the fan. Lift it out.
3. ☐ Loosen the 4 Phillips M2.0×3.0 screws in the order printed on the heat sink, and lift it off.
4. ☐ Clean all old thermal paste (heat-transfer paste) and pads off the processor and the heat sink.
5. ☐ Apply new thermal material (included with the heat sink and board kits), refit the heat sink in order, then the fan.

**Common mistakes**

✗ Reusing old paste.
✗ Using way too much paste.
✗ Tightening one corner all the way first (cracks the chip).

**If something goes wrong**

**Still overheating** → Paste missing or uneven; fan blocked or dead.
**Fan rattles** → Replace the fan.

## Verify before closing the ticket

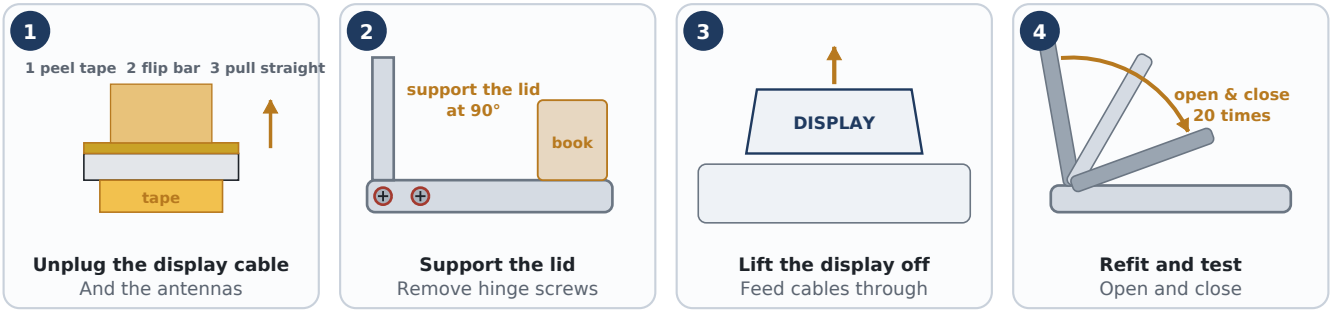
- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

# R9 Display assembly removal (separating lid from base) LEVEL 2

| | |
|---|---|
| **Fixes** | Needed for hinge, lid and some board repairs; replacing a whole display assembly |
| **Part** | Display spared by subcomponent (bezel, panel, cable, camera, hinges, back cover) |
| **Time** | 20–30 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers |

**In plain words.** The display (the whole lid with the screen) is attached to the base by hinges. To take it off you unplug the display cable and antenna wires, hold the lid up, and remove the hinge screws.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

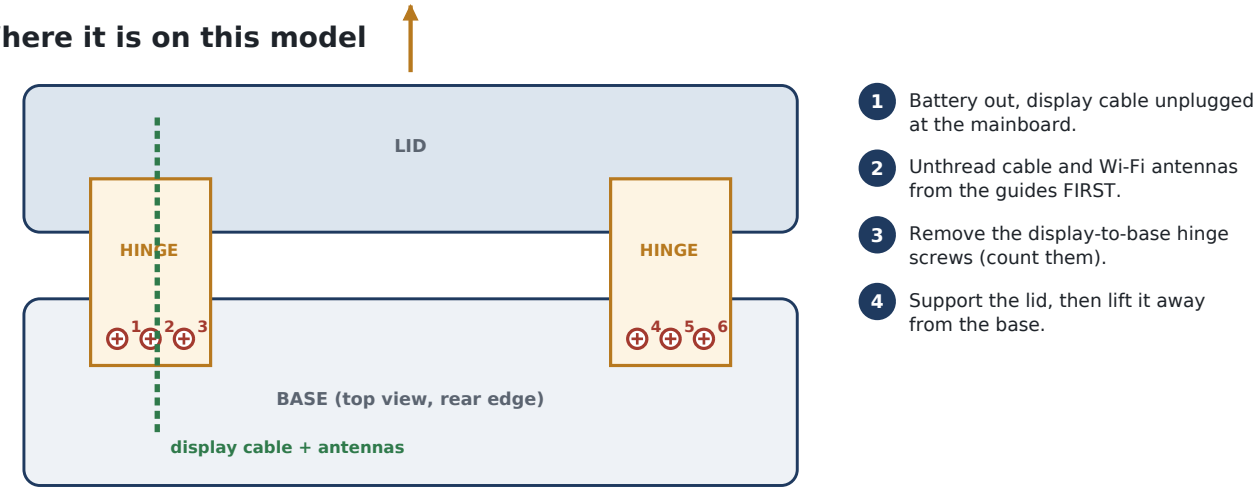



*Illustration: hinge area seen from above with the D cover off. Positions approximate.*

## Step by step

1. ☐ Remove the bottom cover and disconnect the battery. Remove the heat sink first on this model.
2. ☐ Disconnect the display cable and both antennas from the system board; free them from their routing.
3. ☐ Remove the 4 Phillips M2.5×4.5 hinge screws.
4. ☐ Open the display to 90°, support it, and lift it away from the base.

5. ☐ Refit in reverse; route the antennas and display cable back through their clips.

**Common mistakes**
✗ Letting the lid fall when the last screw comes out.
✗ Pinching a cable in the hinge on refit.
✗ Forgetting the antenna wires.

**If something goes wrong**
**Weak Wi-Fi after refit** → Antenna pinched or unplugged.
**Screen flickers** → Display cable not fully seated.

### Verify before closing the ticket

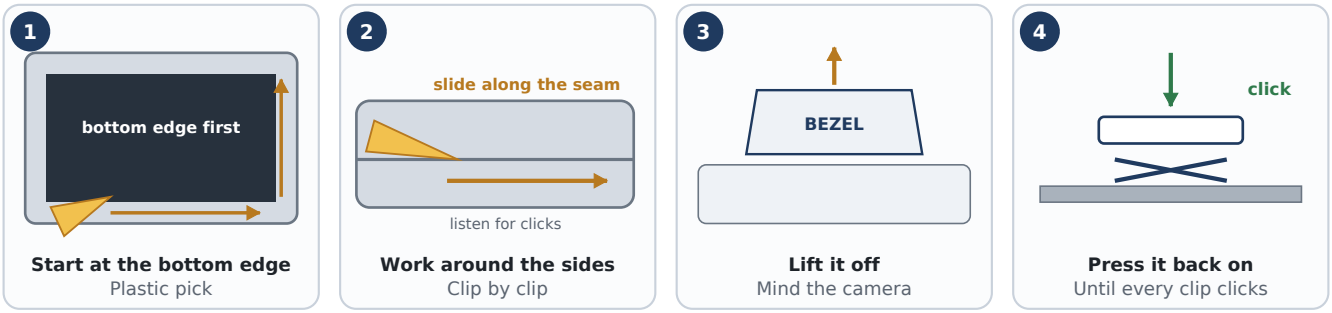
- Display, camera and Wi-Fi all work after reassembly.
- Lid opens through its full range without pinching cables.

## R10 Display bezel replacement — LEVEL 2

| | |
|---|---|
| **Fixes** | Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs |
| **Part** | Display bezel N12925-001; bezel adhesive kit M17753-001 |
| **Time** | 10 minutes |
| **Tools** | Plastic picks, Phillips driver |

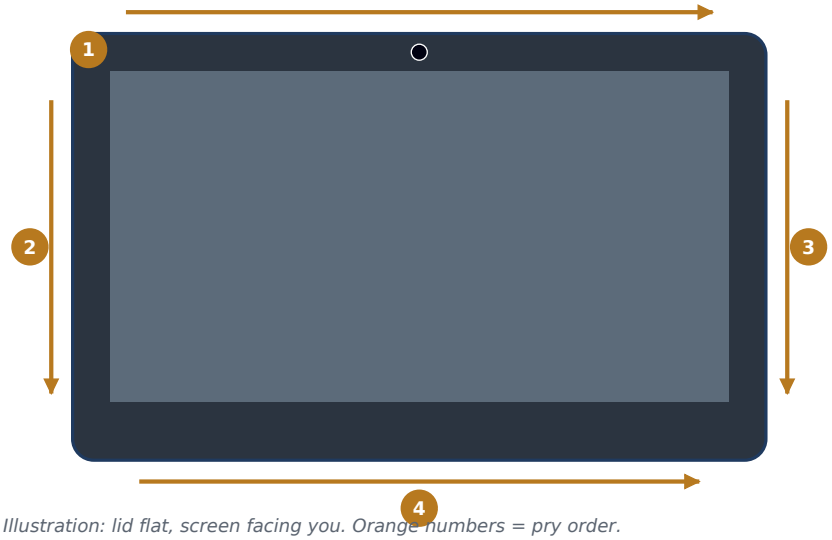
**In plain words.** The bezel is the plastic frame around the screen. It clips (and sometimes glues) onto the lid. You remove it to get to the screen, camera or display cable.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. This bezel is glued as well as clipped.
2. Warm the edge gently (hair dryer on low) if it resists.
3. Slide a thin pick along the adhesive; don't bend the bezel.
4. Fit new adhesive strips on reassembly.
5. Don't pull on the camera window.

*Illustration: lid flat, screen facing you. Orange numbers = pry order.*

## Step by step

1. ☐ Open the lid flat.
2. ☐ The bezel (the plastic frame around the screen) is held with adhesive on all four sides plus hooks on the top and bottom. Warm it gently if it resists.
3. ☐ Work a thin pick under the edge and release it side by side, then unhook the top and bottom.
4. ☐ Clean off old adhesive and fit the adhesive kit (M17753-001) before refitting the bezel.

**Common mistakes**

✗ Prying against the screen (cracks it).
✗ Bending the bezel too far.
✗ Forgetting new adhesive on glued bezels.

**If something goes wrong**

**Gap after refitting** → A clip isn't seated or broke.
**Camera blocked** → Bezel isn't lined up.

## Verify before closing the ticket

- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

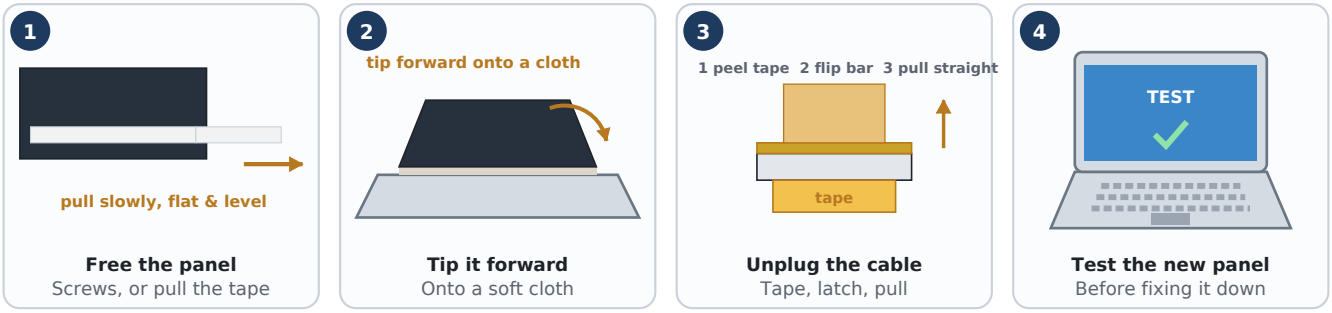
# R11 LCD panel replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Cracked screen, dead pixels, blotches, black screen not fixed by the display cable |
| **Part** | FHD panel N13685-001 (match your label); panel tape kit N13937-001 |
| **Time** | 25–35 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape |

**In plain words.** The LCD panel is the screen itself. Once the bezel is off, it's held by small screws or stretchy tape. It tips forward so you can unplug its cable from the back.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

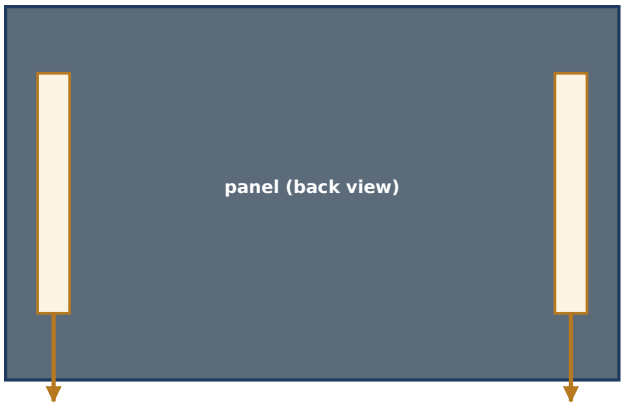



*Illustration: stretch-release panel tape. Tape position illustrative.*

1. Bezel off first. The panel has no screws.
2. Find the stretch-release tape tabs under the left and right edges.
3. Pull each tab slowly, almost flat to the screen, with tweezers. It may take several pulls.
4. Never pry the panel off over unreleased tape; it cracks.
5. Fit new tape strips on reassembly.

**Cracked glass.** Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

## Step by step

1. ☐ Remove the bezel (the plastic frame around the screen).
2. ☐ Pull the stretch-release tape strips out slowly and flat, parallel to the panel.
3. ☐ Tip the panel forward and disconnect the display cable.
4. ☐ Fit new tape (kit N13937-001) before mounting the new panel.

5. ☐ Connect the new panel and test the picture **before** fixing it down.
6. ☐ Refit everything in reverse.

**Common mistakes**
✗ Pressing on the screen face.
✗ Pulling the cable before opening the latch.
✗ Ordering the wrong panel (touch vs non-touch).

**If something goes wrong**
**New panel also black** → Display cable or board: cable repair next.
**Touch not working** → Wrong panel type or touch cable not connected.

### Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

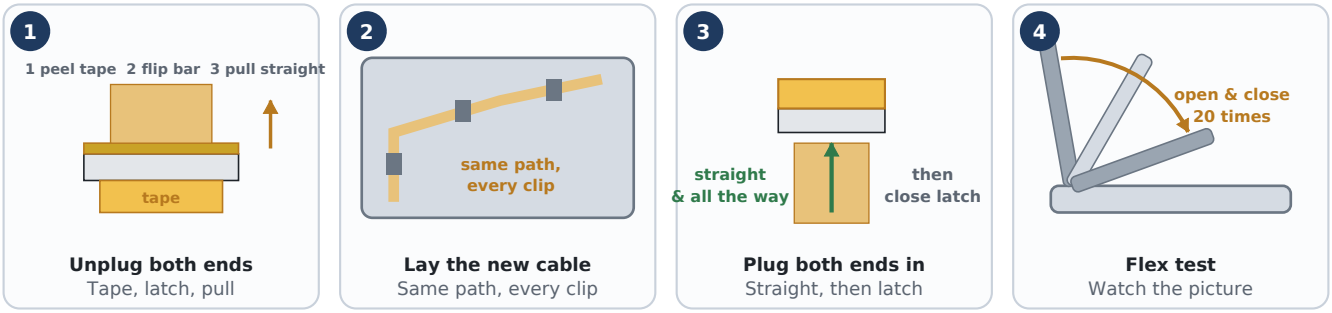
## R12 Display (eDP) cable replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Flicker, lines or black screen that changes with lid angle; external display works |
| **Part** | Display cable N12926-001 |
| **Time** | 40–50 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, Kapton tape |

**In plain words.** The display cable carries the picture from the main board up through the hinge to the screen. Hinges bend it every time the lid moves, so it wears out. Symptoms change when you move the lid.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

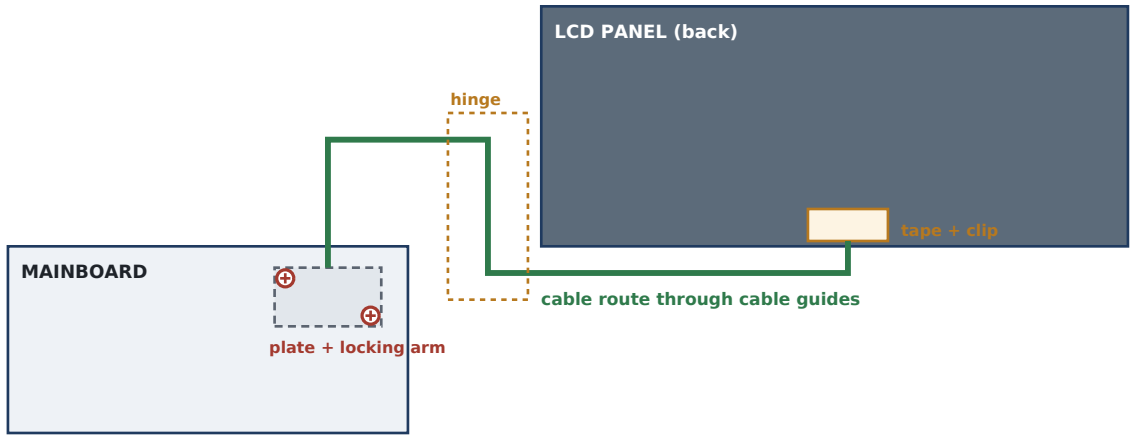



*Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.*

## Step by step

1. ☐ Remove the bezel (the plastic frame around the screen) and panel; disconnect the display cable at the system board.
2. ☐ Free the cable from the back cover and hinge area; note every clip and tape point.
3. ☐ Route the new cable exactly along the old path and reconnect both ends.
4. ☐ Test at every lid angle before reassembly.

**Common mistakes**

✗ Routing the cable a new way (it gets pinched).
✗ Leaving the tape off the connector.
✗ Not testing at different lid angles.

**If something goes wrong**

**Flicker only at some angles** → Cable still pinched or not seated.
**Black screen, external display works** → Panel: LCD panel repair.

## Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

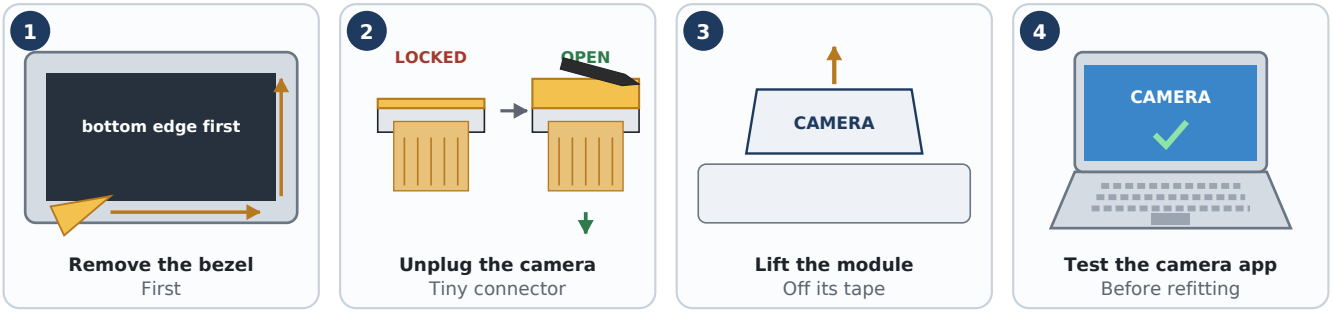
# R13 Camera module replacement

LEVEL 2

| | |
|---|---|
| **Fixes** | Camera not detected, black image, broken lens |
| **Part** | Webcam module N12929-001; webcam cable N10434-001 |
| **Time** | 15–20 minutes |
| **Tools** | Plastic picks, plastic tweezers |

**In plain words.** The camera is a small strip at the top of the screen. It's usually held by sticky tape and connected by a tiny cable. Behind the bezel, it lifts out easily.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




*Illustration: camera at the top-center of the lid. Cap shapes simplified.*

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

## Step by step

1. ☐ Remove the bezel (the plastic frame around the screen).
2. ☐ Remove the 3 Phillips M2.0×2.0 webcam bracket screws.
3. ☐ Disconnect the webcam cable (N10434-001) and lift the module out.
4. ☐ Fit the new module, reconnect, and test before refitting.

**Common mistakes**
✗ Touching the lens.
✗ Forgetting the privacy shutter is closed.
✗ Bending the camera strip.

**If something goes wrong**
**Still not detected** → Check the cable; try S3.
**Image blurry** → Lens dirty or fingerprinted.

### Verify before closing the ticket

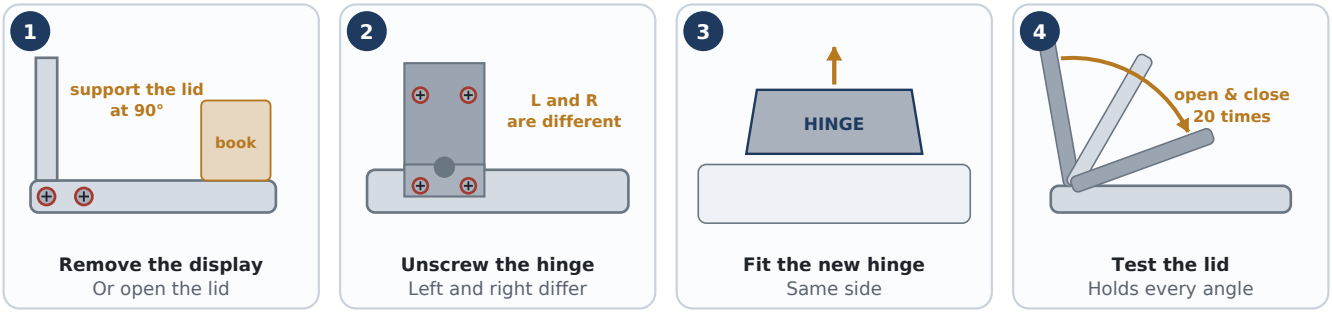
- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

## R14 Hinge replacement — LEVEL 2

| | |
|---|---|
| **Fixes** | Floppy lid, cracked hinge, grinding, lid won't stay up |
| **Part** | Hinges N12927-001; hinge cover N12928-001 |
| **Time** | 30–45 minutes |
| **Tools** | Phillips driver, plastic picks |

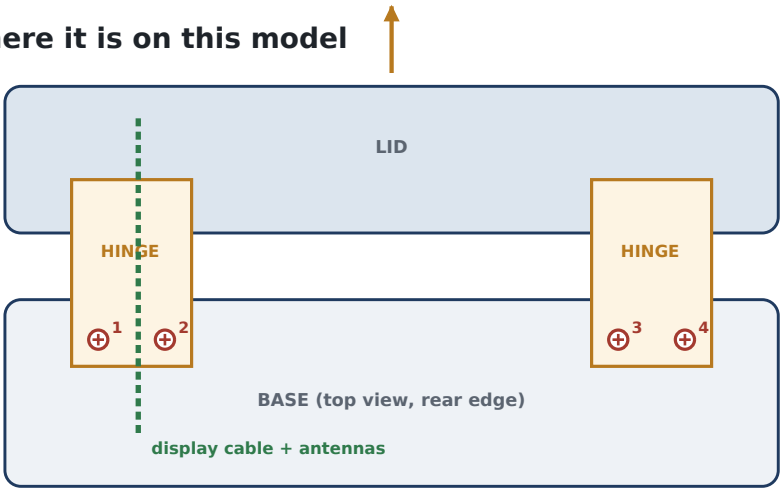
**In plain words.** Hinges are metal brackets that hold the lid to the base and keep it at any angle. A cracked or loose hinge makes the lid floppy.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Base side: 4 M2.5×4.5. Lid side: 3 per hinge.
4. Support the lid, then lift it away from the base.

*Illustration: hinge area seen from above with the D cover off. Positions approximate.*

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

## Step by step

1. ☐ Remove the bottom cover, disconnect the battery, and separate the display assembly from the base.
2. ☐ Remove the hinge cover.
3. ☐ On each hinge remove 2 Phillips M2.5×2.5 broad-head (wide, flat-headed) screws and 1 Phillips M2.5×3.0 screw.
4. ☐ Lift the hinges off the lid.
5. ☐ Fit the new hinges (left and right differ) and reassemble.

**Common mistakes**

✗ Mixing up left and right hinges.
✗ Pinching the display cable in the hinge.
✗ Forcing a stripped screw.

**If something goes wrong**

**Screw spins and never tightens** → Stripped post: cover replacement.
**Lid floppy on one side** → Wrong hinge or loose screw.

## Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

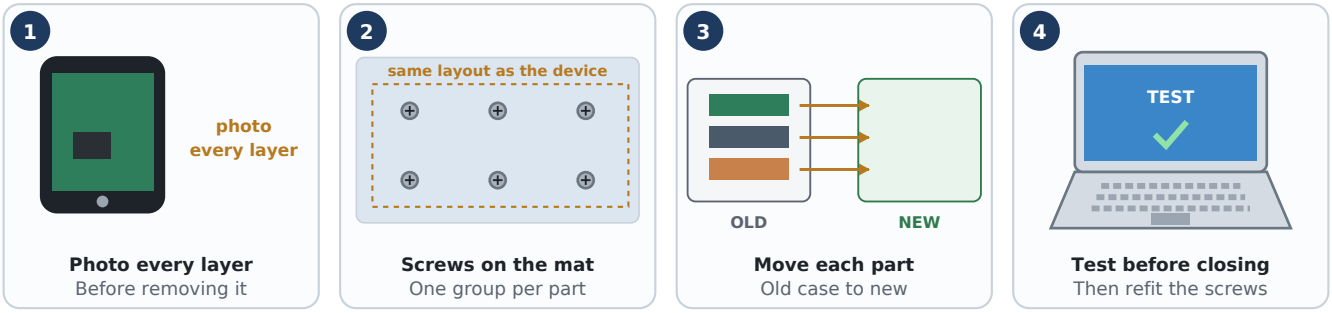
# R15 Keyboard / top cover replacement

LEVEL 3

| | |
|---|---|
| **Fixes** | Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest |
| **Part** | Top cover with keyboard N14927-xxx (by country and backlight; look up by serial) |
| **Time** | 60–90 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape |

**In plain words.** On this model the keyboard is built into the top case (the palm rest). To replace it, every other part moves from the old case to the new one. It's a big job: take photos and go slowly.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




*Illustration: what moves to the new top assembly. Check what comes pre-fitted first.*

The keyboard is built into the top cover. HP's top cover "remains after all other spare parts are removed": this is a full transplant.

## Step by step

1. ☐ Remove the bottom cover and the battery.
2. ☐ Remove every part listed below from the old top cover, in this order: battery, USB board, touchpad and cable, speakers, fingerprint reader and board, fan, heat sink, system board, display assembly.
3. ☐ Check what's pre-fitted on the new top cover, then transfer everything else.
4. ☐ Reconnect every cable, checking against your photos.
5. ☐ Reconnect the battery and test everything before refitting the bottom cover.

**Common mistakes**
✗ Losing track of which screw goes where.
✗ Forgetting a small part (tape, bracket, rubber feet).
✗ Closing up before testing everything.

**If something goes wrong**
**Something doesn't work after** → Check the photo of that area: a cable isn't plugged in.
**Leftover screws** → Find where they go before closing.

## Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

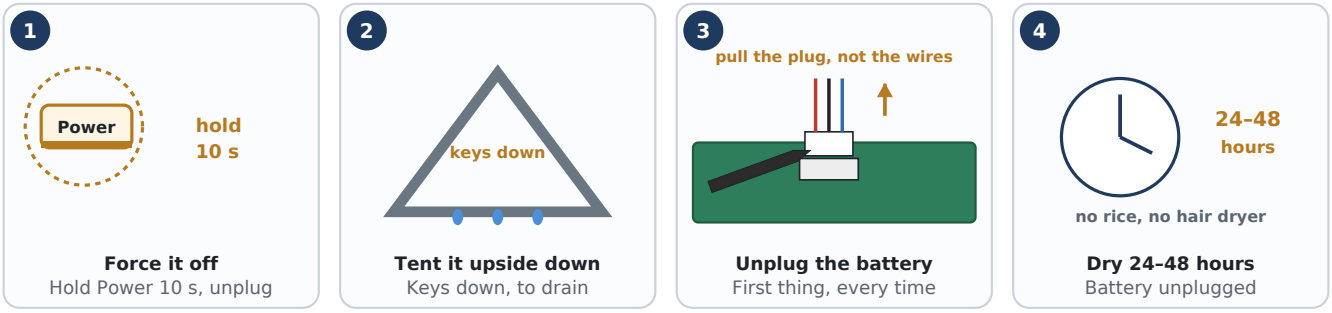
# R16 Liquid spill response

LEVEL 3

| | |
|---|---|
| **Fixes** | Any liquid on or in the device |
| **Part** | Often none; sometimes keyboard/top cover or system board |
| **Time** | 10 min first response; 24–48 h drying |
| **Tools** | 90%+ isopropyl alcohol, soft brush, lint-free cloths |

**In plain words.** Water and electricity don't mix. The goal is to cut the power fast, drain the liquid, and let it dry completely before it's ever turned on again. Speed matters more than anything.

## At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model

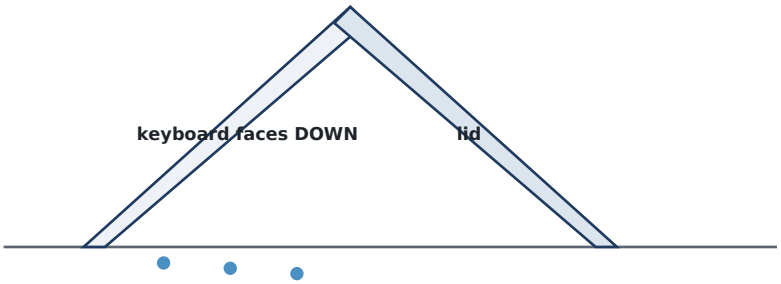



*Illustration: drain position after a spill.*

1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything.

**Do not power on to test.** Current through liquid causes the corrosion that kills boards.

### Step by step

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

**Common mistakes**
✗ Turning it on "just to check".
✗ Using rice or a hair dryer.
✗ Wiping liquid around instead of blotting.

**If something goes wrong**
**White or green crust on the board** → Corrosion: lead tech, Section 6.
**Keys sticky afterwards** → Keyboard repair.

### Verify before closing the ticket

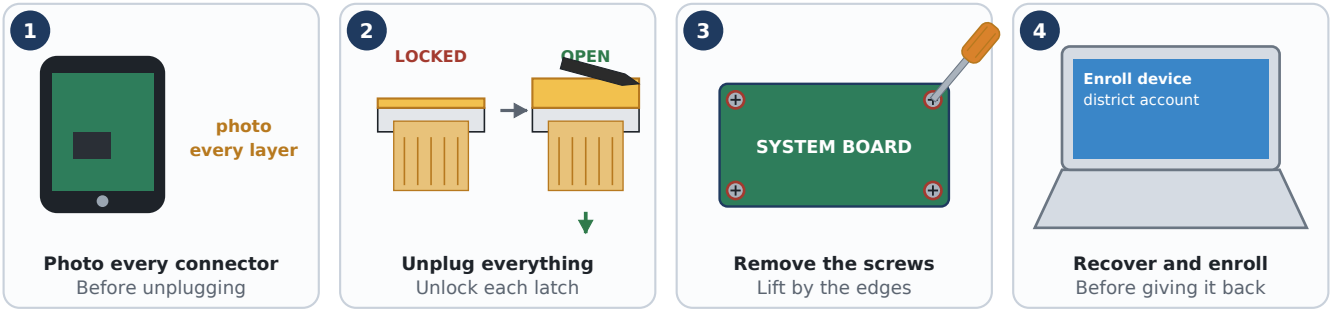
- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

## R17 System board replacement (lead tech) LEVEL 3

| | |
|---|---|
| **Fixes** | Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery |
| **Part** | System board N13676-001 (by configuration; look up by serial) |
| **Time** | 45–90 minutes |
| **Tools** | Phillips driver, plastic spudger, plastic tweezers, ESD strap |

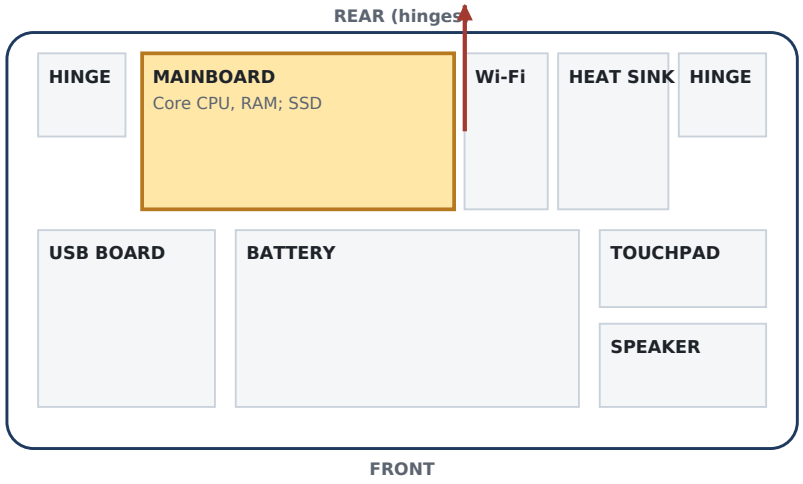
**In plain words.** The system board (main board) is the brain: processor, memory and storage are soldered onto it. Replacing it means unplugging everything, swapping the board, and setting the Chromebook up again. Only after every simpler fix has failed.

### At a glance




The pictures show the idea, not your exact model. The numbered steps below have the details.

## Where it is on this model




1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: display, USB board, touchpad, keyboard, fingerprint, speaker, antennas
3. Lift out after the bracket and all cables are free.
4. Lift by the edges onto the ESD mat.

*Illustration: base interior, mainboard highlighted. Red arrow = lift side.*

**Identity and enrollment.** Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

## Step by step

1. ☐ Remove the bottom cover, the battery, the fan and heat sink.
2. ☐ Lift the Mylar (a thin plastic sheet) and remove the bracket screws: 2 Phillips M2.0×3.0 and 1 Phillips M2.0×2.5.
3. ☐ Disconnect every cable from the board (photograph first).
4. ☐ Remove the 2 Phillips M2.0×2.5 board screws and lift the board out.
5. ☐ Move the SSD to the new board.
6. ☐ Fit the new board, reconnect every cable, refit the parts removed, then run recovery (S3) if needed and enroll.

**Common mistakes**

✗ Replacing the board when a cable was the problem.
✗ Forgetting to move parts that stay (heat sink, SSD).
✗ Not updating the asset record.

**If something goes wrong**

**Still dead with new board** → Check the battery and charger first.
**Won't enroll** → Lead tech checks the Admin console.

## Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

# 6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

| Fault | Why | Route |
|---|---|---|
| USB-C port loose or dead | Soldered to the system board | System board or HP warranty |
| Storage or RAM failure | Soldered (or system-board level) | System board or HP warranty |
| Wi-Fi failure that S3 doesn't fix | No separate Wi-Fi spare listed by HP | HP warranty / system board |
| Burn marks, melted plastic, burnt smell | Fire hazard | Lead tech immediately |
| Any repair needing a soldering iron or hot air | Program policy | HP warranty or lead tech |

## HP support

- **HP support:** support.hp.com
- **HP parts lookup:** partsurfer.hp.com
- **Service guide:** HP Elite c640 14 inch G3 Chromebook Maintenance and Service Guide

# 7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- HP, *Maintenance and Service Guide: HP Elite c640 14 inch G3 Chromebook* (kaas.hpcloud.hp.com, pdf_6048310) — screw counts, sizes, part numbers and procedure order.
- HP PartSurfer (partsurfer.hp.com) — confirm part numbers by serial number.